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Faculty of Engineering

CSE271s: System Dynamics and Control Components
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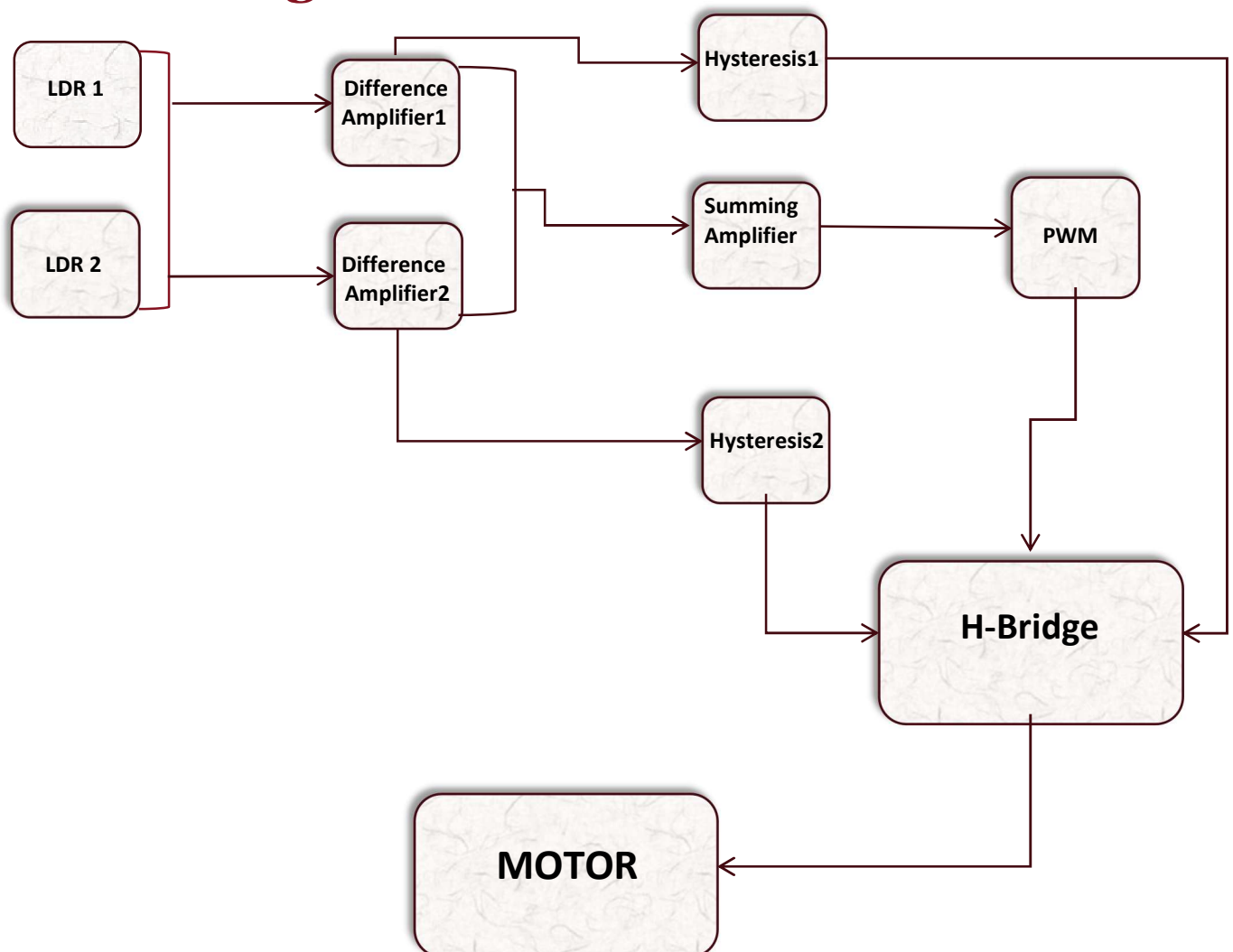
Project Description

This project is a solar tracking system designed to automatically adjust the position of a solar panel toward the sun to maximize energy efficiency by reducing the angle of incidence and increasing power output.

The system uses two LDR sensors to compare light intensity from different directions. Based on the difference, a DC motor rotates the panel left or right toward the stronger light source. To ensure stable operation, a hysteresis control is applied so that the motor stops when the light difference is within a set threshold, preventing unnecessary movement.

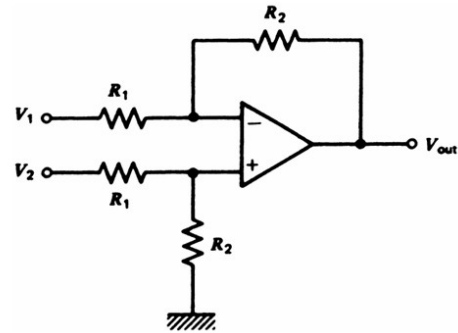
Overall, the system provides a simple and efficient method for improving solar energy collection using automatic light-based tracking.

Block Diagram



2. **Difference amplifier circuit:** A difference amplifier circuit processes the outputs from the LDRs to measure the difference between V_{out} (LDR₁) and V_{out} (LDR₂).

$$V_{out} = \frac{R_2}{R_1} * (V_2 - V_1)$$



Why did we use two difference amplifiers?

- Directional Logic:** The system employs two separate difference amplifiers to independently evaluate the light intensity gradient in both directions.
- Differential Processing:**
 - Difference Amplifier 1: Calculates the potential difference when LDR₁ receives more illumination than LDR₂, triggering the logic for one direction of rotation.
 - Difference Amplifier 2: Calculates the difference when LDR₂ is more illuminated, providing the control signal for the opposite direction.
- Signal Integrity:** Using two amplifiers ensures that the control signals remain unipolar (positive), which simplifying the interface with the H-bridge and the hysteresis stages.
- Speed Control Integration:** The outputs from both amplifiers are fed into a summing stage to provide a consolidated error signal. This signal dictates the duty cycle of the PWM generated by the 555 Timer, ensuring the motor speed is proportional to the light intensity mismatch regardless of the direction.

3. Hysteresis Comparators:

The primary role of the hysteresis stage is to eliminate or rapid, unstable oscillations of the DC motor. This occurs when there is a minimal or zero difference in light intensity between the LDRs due to ambient noise or minor fluctuations.

$$V_H = V_{ref} \quad , \quad V_L = V_{ref} - \frac{R}{R_f} V_0$$

Why did we use two Hysteresis Comparators?

To handle bi-directional nature of solar tracker

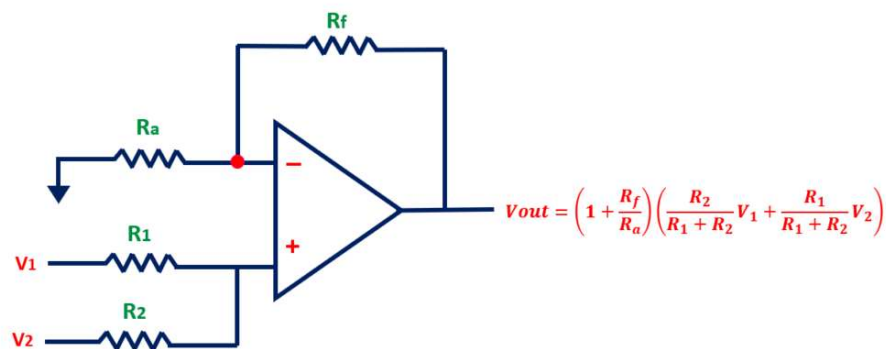
Hysteresis 1: Monitors the output of Difference Amplifier 1. It only triggers the "Right" direction signal once the light imbalance exceeds a predefined upper threshold.

Hysteresis 2: Monitors the output of Difference Amplifier 2. It triggers the "Left" direction signal only when the imbalance in that direction is significant.

System Efficiency: By introducing this "dead band," the system ensures that the motor remains idle until a substantial solar shift is detected, thereby reducing mechanical wear and optimizing power consumption.

4. Summing amplifier:

- **Signal Integration:** The Summing Amplifier acts as a central hub that combines signals from both Difference Amplifiers into a single unified control voltage. This combined voltage is used to drive the PWM stage.
- **Sensitivity Enhancement (Gain):** A gain stage is applied to amplify low-level differential signals, ensuring that even minimal sunlight shifts generate enough torque to overcome mechanical inertia.
- **Dynamic Response:** By increasing the gain, the system becomes more responsive, allowing the solar panel to track the sun with higher precision and speed.



Hand Analysis:

Each LDR (Light Dependent Resistor) is connected in a voltage divider configuration, and the output voltage is taken across each of them.

Voltage across 1st LDR $\rightarrow V_1$

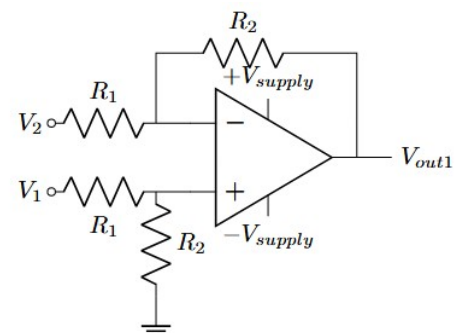
Voltage across 2nd LDR $\rightarrow V_2$

Differential Amplifiers:

We measure the difference in light intensity in terms of the difference between V_1 and V_2 .

1st Differential Amplifier ($V_1 - V_2$):

$$\begin{aligned} V_{out1} &= \frac{R_2}{R_1} (V_+ - V_-) \\ &= \frac{R_2}{R_1} (V_1 - V_2) \end{aligned}$$



We connected the positive supply to V_{cc} and the negative supply to the ground.

$$R_1 = 1 \text{ k}\Omega, R_2 = 2 \text{ k}\Omega$$

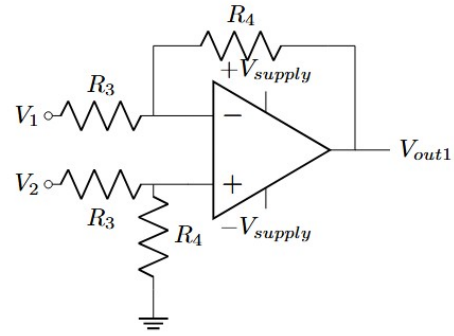
$$\therefore V_{out1} = 2(V_1 - V_2)$$

2nd Differential Amplifier ($V_2 - V_1$):

Same connections as the 1st Differential Amplifier just switching input terminals.

$$V_{out2} = \frac{R_4}{R_3}(V_2 - V_1) = 2(V_2 - V_1)$$

The motor needs to move in 2 directions; thus, we designed the circuit with 2 differential amplifiers whereas only one of them gives an output at a time. For example, if $V_1 > V_2 \rightarrow$ 1st amplifier outputs a positive value while the 2nd outputs zero.



Hysteresis Comparator:

To ensure the motor doesn't jitter or lag, hysteresis comparators are added to the outputs of both differential amplifiers.

We will make the hysteresis window = 0.9 V (around V_{ref}).

$$V_{ref} = V_{cc} \times \frac{6k\Omega}{10k\Omega + 6k\Omega} = 9 \times \frac{6}{16} \approx 3.4V$$

$$V_H = V_{ref} = 3.4V, V_L = V_{ref} - \frac{R}{R_f}V_0 = 3.4 - (0.1)(9) = 2.5V$$

$$V_H - V_L = \frac{R}{R_f}(V_{OH} - V_{OL})$$

Assuming window 0.9V and $(V_{OH} - V_{OL}) = 9V$:

$$\frac{R}{R_f} = \frac{0.9}{9} = \frac{1}{10} \Rightarrow R = 10k\Omega, R_f = 100k\Omega$$

Same values were selected for the 2nd hysteresis comparator.

Summing Amplifier:

Used to get the absolute difference between V_1 and V_2 by connecting to the outputs of the two differential amplifiers.

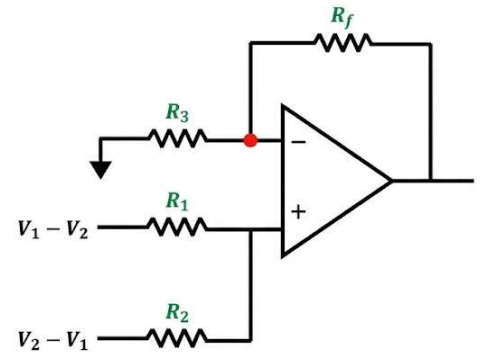
$$A_v = 1 + \frac{R_f}{R_3} \rightarrow \frac{R_f}{R_3} = A_v - 1$$

$$V_+ = \frac{V_a/R_1 + V_b/R_2}{1/R_1 + 1/R_2}$$

Assuming $R_1 = R_2 \rightarrow V_+ = \frac{V_a + V_b}{2}$ ($V_a = V_1 - V_2, V_b = V_2 - V_1$)

$\therefore$ only one of V_a, V_b active

$\therefore V_+ = \frac{V_a}{2}$ or $\frac{V_b}{2}$ (depending on light direction)



If we assumed that V_a is the one that has a value:

$$V_{out} = \left(1 + \frac{R_f}{R_3}\right)V_+ = \left(1 + \frac{R_f}{R_3}\right)\left(\frac{V_a}{2}\right)$$

We set R_a with a potentiometer to control sensitivity to ambient light.

Part (B): Control Stage

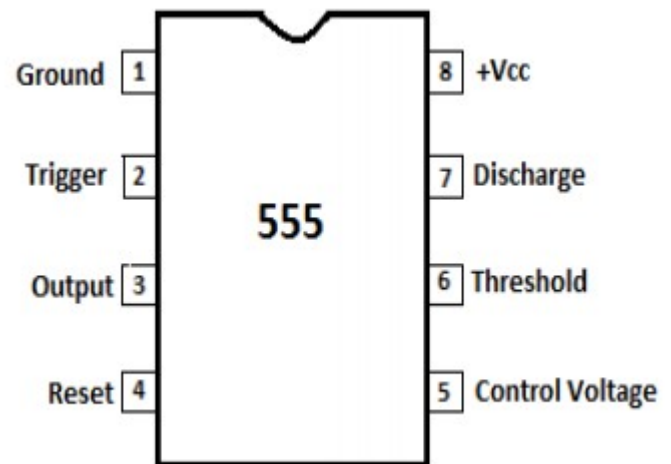
Objective:

The control stage translates the conditioned error signal into physical motion through two primary sub-circuits:

Pulse Width Modulation (PWM) Generation:

Dual 555 Timer PWM Configuration: The speed control mechanism in this project is implemented through a dual-stage 555 Timer configuration to generate a high-precision Pulse Width Modulation (PWM) signal.

Continuous Triggering (Astable Mode): The primary 555 Timer operates in Astable Mode, acting as a free-running oscillator. Its sole purpose is to provide a continuous, fixed-frequency square wave that serves as a periodic trigger for the subsequent stage.



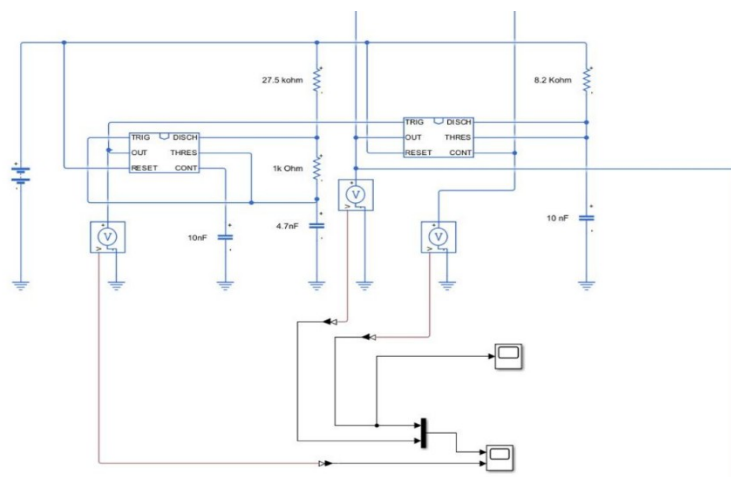
Variable Pulse Generation (Monostable

Mode): The secondary 555 Timer is configured in Monostable Mode. Instead of a fixed pulse, it responds to each trigger from the first timer by generating a single pulse.

Dynamic Control via the Control Pin: The innovation in this setup lies in the use of the Control Pin (Pin 5) of the second timer. By feeding the output voltage from the Summing Amplifier into this pin, the internal threshold levels are dynamically adjusted.

Speed Modulation: This adjustment allows the system to vary the Duty Cycle of the output signal in real-time. Consequently, the DC motor's speed becomes directly proportional to the sunlight intensity mismatch detected by the LDRs.

Circuit Design:



Hand Analysis:

Astable Timer Calculations:

$$T_L = \ln(2) C R_B, T_H = \ln(2) C(R_A + R_B)$$

$$T = T_H + T_L, f = \frac{1}{T} = \frac{1}{\ln(2) C(2R_B + R_A)}$$

Calculations for 10kHz:

$$\frac{1}{\ln(2) C(2R_B + R_A)} = 10^4$$

With $C = 4.7 \text{ nF}$, $R_B = 1 \text{ k}\Omega$, $R_A = 28.7 \text{ k}\Omega$:

$$T_L = 3.26 \text{ }\mu\text{sec}, T_H = 96.76 \text{ }\mu\text{sec}$$

$$T = 100.02 \text{ }\mu\text{sec} \Rightarrow f \approx 10.41 \text{ kHz}$$

Monostable Timer Calculations:

$$T_w = C_m R_m \ln \left(\frac{1}{1 - \frac{V_{control}}{V_{cc}}} \right)$$

$$\text{Duty Cycle} = \frac{T_w}{T} = T_w \cdot f_{astable}$$

$$\text{Constants} \Rightarrow 10.410 \times 10^3 \times 10 \times 10^{-9} \times 8.2 \times 10^3 \approx 0.85362$$

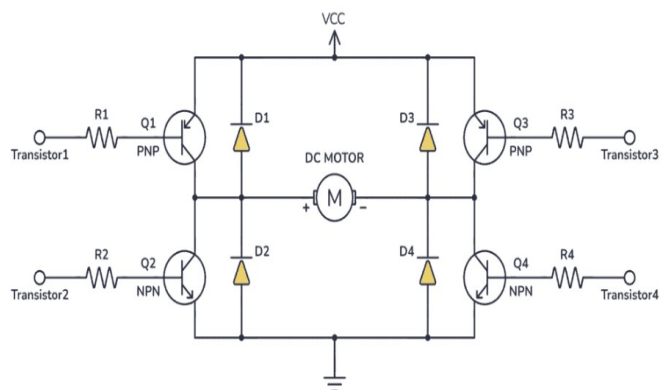
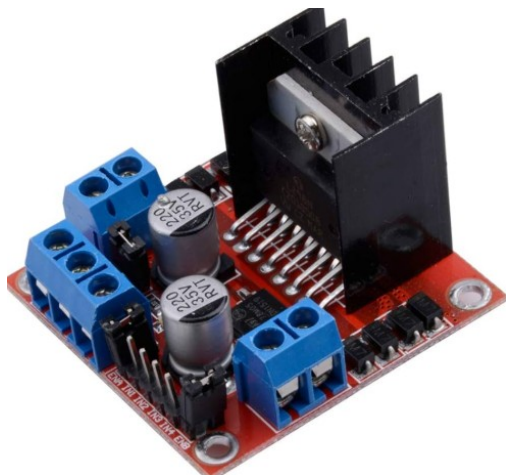
We chose:

$$R_m = 2 \text{ k}\Omega, C_m = 10 \text{ nF}$$

Part (C): H-Bridge and DC Motor

H-Bridge:

The H-Bridge is the power interface responsible for determining the rotational direction of the DC motor. It consists of four electronic switches arranged in an "H" configuration, enabling the system to reverse the polarity across the motor terminals.



Directional Logic:

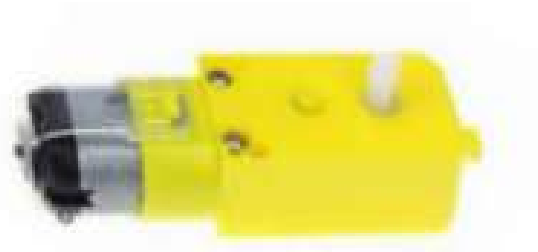
- **First Direction:** By activating the first diagonal pair of switches (S1 and S4), current flows through the motor in one direction, initiating a forward movement.
- **Second Direction:** Activating the opposite diagonal pair (S2 and S3) reverses the current path, causing the motor to rotate in the opposite direction.
- **PWM Integration:** In this design, the H-Bridge does not just switch the motor ON or OFF; it acts as a gate for the PWM signal. By feeding the 555-timer's output into the H-bridge's enable or switching pins, the circuit delivers varying amounts of average power to the motor, effectively controlling its speed while maintaining torque.

The Actuator (DC Motor):

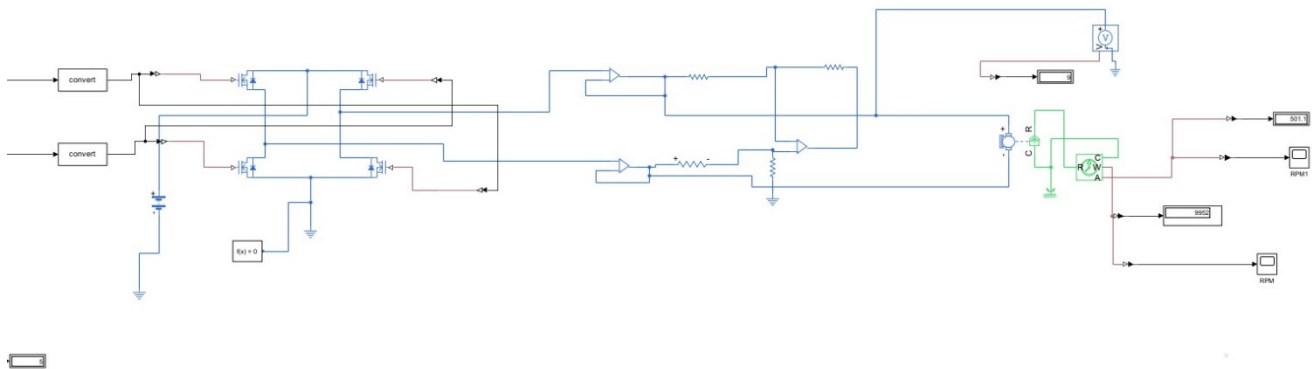
The DC motor is the final part of the system that physically moves the solar panel. Its operation depends on the signals received from the control stage:

Movement and Direction: The motor rotates based on the H-Bridge setting. This allows the panel to follow the sun's path throughout the day.

Speed Control: The motor does not just turn on and off; its speed changes dynamically. By using the PWM signal, the motor moves faster when the sun is far away and slows down as the panel reaches the correct position.



Circuit Design:

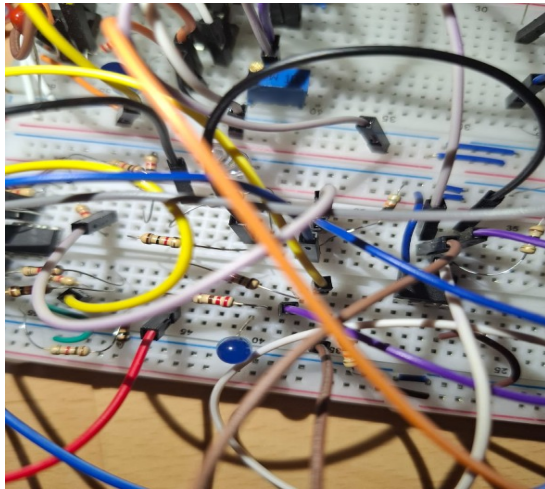


Components List

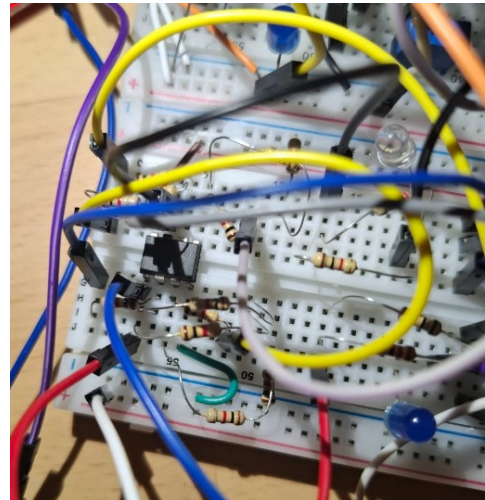
Component	Quantity	Price (EGP)
LM358(op-amp)	4	$10 \times 4 = 40$
LDR	2	$20 \times 2 = 40$
Adapter 9V	1	150
2-Input AND Gate	1	15
DC Barrel Jack Adapter	1	15
NE555	2	$3.75 \times 2 = 7.5$
Motor Drive Module	1	80
Potentiometer (50 k Ω)	1	10
Potentiometer (10 k Ω)	2	$10 \times 2 = 20$
Breadboard	3	$60 \times 3 = 180$
Jumper wires	-	100
Capacitor 10 nF	1	0.52
Capacitor 4.7 nF	1	3
Capacitor 100 nF	1	0.52
Resistances (1k, 10k, 3k, 100k, 220)	30	$0.5 \times 30 = 15$
Total		676.5

Circuit Implementation

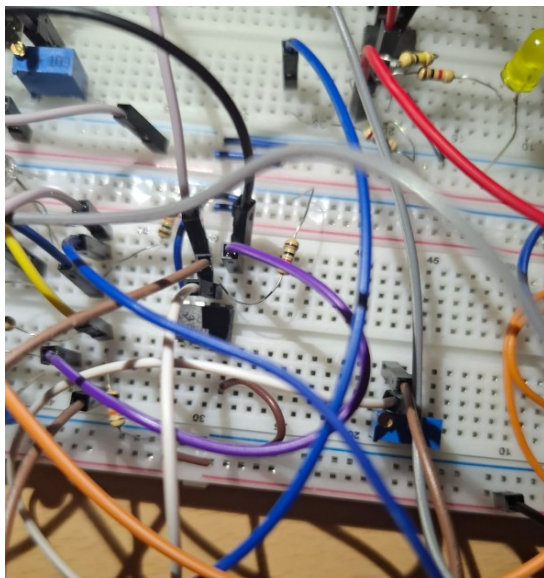
1) Voltage divider



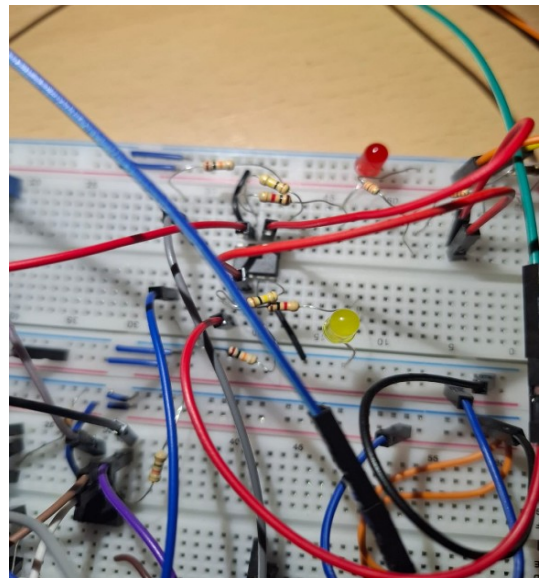
2) Difference Amplifier



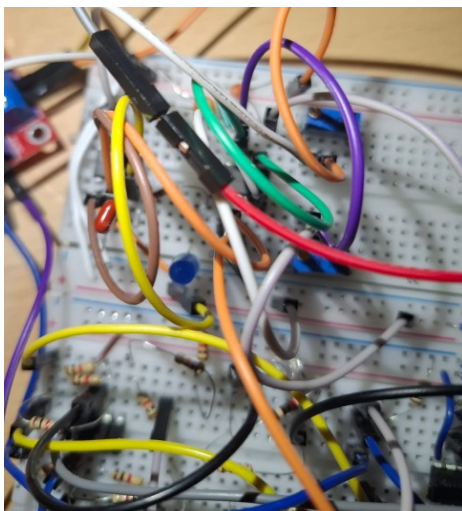
3) Summing Amplifier



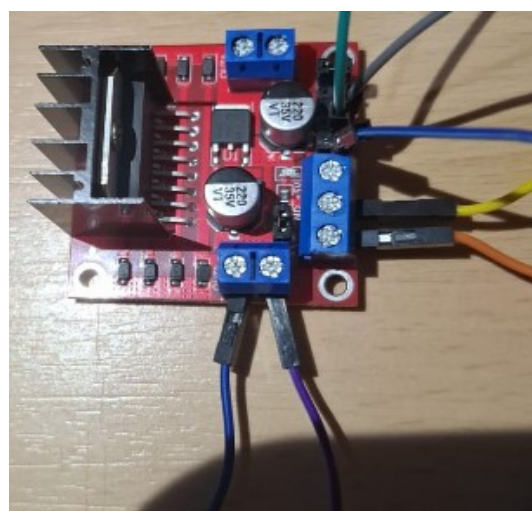
4) Hysteresis



5) PWM



6) H-Bridge



Project Challenges & Technical Hurdles

1. Evolution of the Signal Conditioning Architecture:

Our initial design utilized a standard differential amplifier followed by a precision rectifier to obtain an absolute value of the light intensity mismatch. However, this configuration presented significant implementation challenges:

- **Power Supply Constraints:** The precision amplifier required a stable dual-rail supply ($+V_{cc}$ and $-V_{cc}$) to process bipolar signals accurately, which increased the complexity of our power management stage.
- **Design Pivot:** To streamline the system, we transitioned to using two separate difference amplifiers. This allowed us to calculate directional gradients independently, ensuring all control signals remained unipolar (positive). This change eliminated the need for a negative supply and simplified the interface with the H-bridge and hysteresis stages.

2. Mitigating System Instability and Jitter:

A major challenge in solar tracking is the "jitter" caused by ambient noise or minute fluctuations in light intensity. Without a buffer zone, the DC motor would attempt to react to negligible changes, leading to rapid, unstable oscillations and unnecessary mechanical wear. Thus, the implementation of hysteresis comparators was vital. By establishing a "dead band" through predefined upper and lower thresholds, we ensured the motor remains idle until a substantial solar shift occurs. This stabilized the operation and significantly improved system efficiency.

3. Simulation vs. Real-World Hardware Discrepancies:

Bridging the gap between the MATLAB simulations and the physical breadboard implementation revealed critical discrepancies:

- **Parameter Uncertainty:** In simulation, motor models often assume ideal conditions. In practice, the internal resistance (R) and capacitance (C) of the physical motor and the 555-timer timing components were difficult to characterize precisely.
- **Circuit Interference:** The inductive loads from the real-world motor caused feedback issues in the timer circuit that were not fully captured in the digital environment, requiring iterative manual tuning of the PWM frequency settings.

4. Balancing Sensitivity and Actuator Response:

Achieving a balance between the sensitivity of the LDR sensors and the physical response of the motor was a complex tuning task.

Drive link

<https://drive.google.com/drive/folders/1FdT9wrXnSGw9Yavum2D9WfNMIDF8TBv4?usp=sharing>

Team Members Contribution

Noura Muhammed Ezzat Muhammed	-Signal conditioning Simulation -Circuit implementation -Maket implementation -Report Documentation
Nadine Nader Mohamed Amin	-Signal conditioning Simulation -Circuit implementation -Maket implementation -Purchase the component
Nour Ahmed Mohamed Seddik	-H-Bridge & Motor Simulation -Circuit implementation -Signal Conditioning Hand Analysis
Hadeer Hesham Ahmed	-Signal conditioning Simulation -Circuit implementation
Jana Mohamed Ahmed	-Timer Simulation -Circuit implementation
Salma Sameh Batal Abdelhalim	-Timer Simulation -Circuit implementation -Purchase the component -Report Documentation
Mariam Osama Mahmoud Elkady	-H-Bridge & Motor Simulation -Circuit implementation -Purchase the component
Marwa Osama Mahmoud Elkady	-H-Bridge & Motor Simulation -Circuit implementation -Purchase the component
Hana Hassan Sayed Hassan	-Timer Simulation -Circuit implementation
Shorouk Amr Mohamed	-Timer Simulation -Circuit implementation -Timer Hand Analysis -Purchase the Components